



Recommended Capstone: "Intelligent Loan Refinancing Recommender System"

"Predicting Refinancing Benefit and Recommending Optimal Loan Products Using ML + Explainable AI"

This is the **perfect capstone for you** — it directly mirrors what Unlend.in does, satisfies every capstone requirement, demonstrates a broad AI skill set, and becomes a genuine portfolio centerpiece you can show both academic evaluators and AI job interviewers.

Why This Beats Generic Projects

Most capstone projects are just classification exercises on tired datasets. Yours will tell a **business story**: *"Given a borrower's current loan situation, should they refinance? Into what product? And how much will they save?"* — This is a multi-layered AI problem, not a one-trick model.

The Dataset Stack (all free, publicly available)

You'll layer **two datasets** to create a richer problem:

Primary: LendingClub Loan Dataset (Kaggle) The gold standard for personal loan ML projects — 2.2 million loan records (2007–2018) with 74+ features including loan amount, interest rate, purpose, DTI ratio, credit grade, employment info, payment history, and loan status (fully paid / charged off). It even includes a **purpose** field with values like **debt_consolidation** and **credit_card** — directly mapping to what Unlend does.

Secondary: Home Credit Default Risk Dataset (Kaggle) ~350,000 loan applications with rich borrower profiles — income, credit bureau data, previous loan behavior. Great for the credit risk scoring component.

Bonus enrichment (optional, for home loan angle): Freddie Mac Single Family Loan dataset (public, US-based but structurally identical to Indian home loans) — adds a mortgage refinancing dimension.

What You'll Build: A 3-Layer System

Layer 1 — Credit Risk Scoring Model

"Is this borrower creditworthy enough to be refinanced?"

- Features: DTI, credit grade, employment length, income, delinquency history
- Models: Logistic Regression (baseline) → XGBoost / LightGBM (final)
- Output: Probability of default (risk score)
- Techniques: SMOTE for class imbalance, cross-validation, ROC-AUC evaluation

Layer 2 — Refinancing Benefit Prediction Model

"How much will this borrower save if they refinance?"

- Engineer a target variable: **estimated monthly savings** based on current rate vs. predicted new rate
- Models: Gradient Boosted Regression (XGBoost) to predict new eligible interest rate
- Segment by loan type: personal loan, credit card → personal loan, home loan
- This is the core supervised learning component and the heart of your capstone

Layer 3 — Refinancing Recommendation Engine

"Which loan product should we recommend, and why?"

- Rule-based + ML hybrid: combine the risk score and savings estimate with loan product eligibility rules
- Use **SHAP values** to explain every recommendation (why does this borrower qualify? which factors helped/hurt?)
- Build a simple decision matrix: High Savings + Low Risk = Strong Recommend; Low Savings + High Risk = Do Not Recommend
- Optionally: add a clustering step (K-Means / DBSCAN) to segment borrowers into refinancing personas

AI Technologies You'll Demonstrate

Technology	Where It Appears
Supervised Learning (Classification)	Credit risk model — XGBoost, Logistic Regression
Supervised Learning (Regression)	Interest rate / savings prediction

Unsupervised Learning (Clustering)	Borrower persona segmentation
Feature Engineering	DTI ratios, savings delta, credit utilization
Class Imbalance Handling	SMOTE, class weighting
Explainable AI (SHAP)	Per-borrower recommendation explanation
Data Pipeline / EDA	Pandas, Seaborn, Matplotlib
Model Evaluation	ROC-AUC, RMSE, confusion matrix, lift curves
Recommendation System Logic	Rule + ML hybrid engine
Ethical/Responsible AI	Bias analysis by income group, geography

This is **9–10 distinct AI skills** in one project — far more than any typical capstone.

How It Maps to the Capstone Timeline

Module 11 Problem Statement: *"Can we predict which borrowers will benefit from loan refinancing and recommend optimal products using credit risk and savings prediction models?"*

Module 16 Formal Proposal: Research question + LendingClub/Home Credit datasets + XGBoost + SHAP methodology

Module 20 EDA + Initial Report: Full EDA of borrower profiles, interest rate distributions, loan purpose breakdown, early model results

Module 24 Non-Technical Report: Frame it as a business story — *"For every 1,000 borrowers analyzed, our model identifies X who could save ₹Y per month through refinancing"*

The Portfolio / AI Job Angle

This project lets you demonstrate things that are **specifically hot in AI hiring right now**:

1. **End-to-end ML thinking** — not just fitting a model, but designing a system with a business outcome
2. **Explainability (XAI)** — SHAP is explicitly requested by most fintech AI teams
3. **Domain expertise + AI** — you're a CTO in fintech building an AI system for fintech; that's a powerful narrative

4. **Responsible AI** — showing bias analysis across income groups signals seniority

You can present this as *"I built a production-grade refinancing recommendation engine as my capstone, which mirrors the core AI problem at Unlend.in"* — that story works both in academic evaluation and in any AI/ML job interview.

One Enhancement (High Impact, Low Effort)

Integrate this in unlend, where you scan the user's credit report, or a user inputs their loan details and gets a real-time refinancing recommendation with an explanation. This makes it interactive and demonstrable in interviews — a massive differentiator. You'd have it live on HuggingFace Spaces or Render for free.